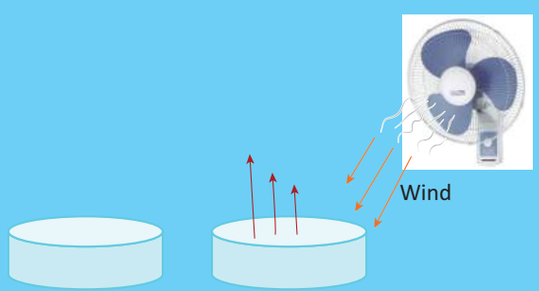
ACTIVITY

Activity 2

You must be remembering an activity done in Class six, in which we have taken two same shaped glasses and fill them with equal amount of water from same tap. We kept one under the hot sun and other under the shadow. After three to four hours, we saw that there is difference in water levels. The one kept in the hot place witness more evaporation compared to the one in shade. From this we can conclude that higher the temperature, the rate of evaporation will be more. As the temperature increases, more molecules are able to break free from the surface. Thus the rate of evaporation increases with rising temperature.

Activity 3

Take two pans, one wide and another narrow. Fill hot water in both to the same depth. Keep them in open. Observe after one to two hours. The pan that is wide has cooled more than the narrow one. That is more the surface area; the rate of evaporation is more.



From this, can you guess why we unfurl the clothes while putting them to dry, rather than just drape them over the cloth line?

Greater the surface of conversion of a liquid, more molecules are available for evaporation.

Activity 4

Take sugar solution in a shallow, broad bowl. Place the bowl in hot sun for a few hours. See that the bowl does not get any disturbance for the whole day. You can see that the solvent in the sugar solution evaporates leaving the sugar crystals in the bowl.

Evaporation is a slow process and occurs only at the surface of the liquid.

3.3.3 Freezing

Water in the freezer compartment of a refrigerator gets cooled and solidifies to form ice. In this case, the liquid water changes into solid water called ice.

Only a change in state (from liquid to solid) takes place during the freezing of water to form ice, but no new substance is formed. So, the freezing of water is a physical change.

Upon cooling a liquid, the particles loose energy and vibrate less vigorously. When the particles possess less energy, they can experience strong forces of attraction between one another. The particles move closer to each other and movement of particles is also restricted. For example, when liquid water is cooled to 0° C, it freezes to become ice. Freezing occurs when the freezing point is reached. The liquid changes to its solid state.

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